SHIN, Young Ah

Seoul, Republic of Korea | yashin95@korea.ac.kr
Homepage | LinkedIn | Google Scholar | ORCID | GitHub

RESEARCH INTERESTS

My research focuses on applied cryptography and privacy-enhancing technologies for building trustworthy distributed and autonomous systems.

– Applied cryptography and provably secure protocols
– Secure delegation and accountability for autonomous AI agents
– Privacy-preserving payment and blockchain protocols
– Privacy-enhancing technologies and federated learning
– Lightweight cryptographic protocols for mobile and IoT systems

EDUCATION

Ph.D. in Cybersecurity, Korea University, School of Cybersecurity *Mar. 2020 – Aug. 2026*

Privacy-Enhancing Technologies (PET) Lab. | Advisor: Prof. Ik Rae Jeong

Dissertation: *Practical and Secure Online and Offline Cryptographic Protocols for Multi-User and Multi-Device Settings in Mobile Applications*

– Designed practical, provably secure identity-based multi-proxy signature and on/offline signcryption protocols for multi-user and multi-device mobile environments.
– Formalized memory-corruption threats between offline and online phases and proposed constructions with provable security against the resulting attacks.

B.S. in Convergence Security Engineering, Sungshin Women's University *Mar. 2016 – Feb. 2020*

Thesis: *Blockchain-Based Mobile Coupon Trading Platform for Campus and Retailers Using zk-SNARKs*

PROFESSIONAL POSITIONS

Research Professor, School of Cybersecurity, Korea University *Sep. 2026 – Aug. 2027*

Manage research funding and performance reporting for the research project *Master's and Doctoral Program for Advanced Talent Development in Personal Data Protection and Responsible Use*, funded by Korea's Personal Information Protection Commission (PIPC).

Adjunct Professor, Department of Bigdata & Information Security,
Seoul Cyber University *Jun. 2026 – Aug. 2027*

Teach two English-medium undergraduate courses, *Introduction to Cybersecurity* and *AI Programming Fundamentals*.

Lecturer, Graduate School of Privacy & Data Protection, Korea University *Mar. 2025 – Feb. 2027*

Teach two Korean-medium graduate courses, *Privacy Enhancing Technology* and *Privacy by Design*, for working professionals.

TEACHING EXPERIENCE

Introduction to Cybersecurity, Seoul Cyber University *Fall 2026 (Scheduled)*

Department of Bigdata & Information Security | English | Online | 3 credits.

AI Programming Fundamentals, Seoul Cyber University *Fall 2026 (Scheduled)*

Department of Bigdata & Information Security | English | Online | 3 credits.

PDP008: Privacy by Design, Korea University *Mar. 2025 – Present*

Graduate School of Privacy & Data Protection | Korean | Offline | 2 credits.

PDP003: Privacy Enhancing Technology, Korea University *Mar. 2025 – Present*

Graduate School of Privacy & Data Protection | Korean | Offline | 2 credits.

PUBLICATIONS

Journals

- **Young Ah Shin**, Geontae Noh, and Ji Young Chun, “A Survey of Leader Election Techniques without Oracles in Public Blockchain Systems,” *Journal of Future Society*, vol.16, no. 2, pp. 170-185, 2025. (in Korean) doi:10.22987/jifso.2025.16.2.170
- Gyeong Min Baek, **Young Ah Shin**, Gye Hyun Jang, and Ik Rae Jeong, “Block Number-Based Address Clustering for Bitcoin Taproot Upgrade,” *IEEE Access*, vol. 13, pp. 94523–94537, 2025. doi:10.1109/ACCESS.2025.3570459
- Min Seob Lee, **Young Ah Shin,** Ji Young Chun, “Systematic Research on Privacy-Preserving Distributed Machine Learning,” *KIPS Transactions on Computer and Communication Systems*, vol. 13, no. 2, pp. 76-90, 2024. (in Korean) doi:10.3745/TKIPS.2024.13.2.76
- Kyoung Jae Jung, **Young Ah Shin**, Geontae Noh, Ji Young Chun, and Ik Rae Jeong, “Secure Medical Data Sharing Techniques that Can Be Utilized in IoT Environments,” *Journal of the Korea Institute of Information Security & Cryptology*, vol. 34, no. 6, pp. 1369-1383, 2024. (in Korean) doi:10.13089/JKIISC.2024.34.6.1369
- **Young Ah Shin**, Ji Young Chun, and Geontae Noh, “Sustainable Blockchain-Based Multi-Party Micropayment System,” *Journal of Future Society*, vol.15, no. 3, pp. 231-244, 2024. (in Korean) doi:10.22987/jifso.2024.15.3.231
- **Young Ah Shin**, Ik Rae Jeong, and Jin Wook Byun, “Identity-Based Multiproxy Signature with Proxy Signing Key for Internet of Drones,” *IEEE Internet of Things Journal*, vol. 11, no. 3, pp. 4191–4205, 2024. doi:10.1109/JIOT.2023.3300299
- **Young Ah Shin** and Ji Young Chun, “Secure Cascade Vertical Federated Learning over Distributed Labels Against Passive Label Inference Attack,” *Asia-pacific Journal of Convergent Research Interchange,* vol.9 no. 11, pp. 47-56, 2023. (in Korean) doi:10.47116/apjcri.2023.11.05
- **Young Ah Shin**, JoonSub Kim, Ik Rae Jeong, and Geontae Noh, “Secure and Scalable Multi-Party Payment Channel Hub in Blockchain,” *Asia-pacific Journal of Convergent Research Interchange*, vol. 9, no. 11, pp. 57-67, 2023. (in Korean) doi:10.47116/apjcri.2023.11.06
- **Young Ah Shin**, Geontae Noh, Ik Rae Jeong, and Ji Young Chun, “Securing a Local Training Dataset Size in Federated Learning,” *IEEE Access*, vol. 10, pp. 104135–104143, 2022. doi:10.1109/ACCESS.2022.3210702

Manuscripts Under Review

- **Young Ah Shin**, Ik Rae Jeong, and Jin Wook Byun, “Secure ID-Based On/Offline Signcryption from Online and Offline Attacks.” Submitted Q1 2026.
- Joon Ho Lim, **Young Ah Shin**, and Ik Rae Jeong, “PETRA: Privacy-Preserving Efficient Tracing for Regulation-Compliant CBDC.” *IEEE Transactions on Information Forensics and Security*; submitted Q4 2025.

Conference (Korean/Oral)

- Kyoung Jae Jung, Young Ah Shin, Ik Rae Jeong, and Ji Young Chun, “Comparative Analysis of Computation and Time Efficiency by Participant type in a Homomorphic Encryption-based Federated Learning Environment,” *Conference on Information Security and Cryptography-Winter(CISC-W)*, 2023.
- JoonSub Kim, **Young Ah Shin**, and Ik Rae Jeong, "Performance Analysis of Key Agreement Protocols in IoD Environments for Secure Data Communication with Data Collection Drones," in *Proc. Summer Annual Conference of the ICT Platform Society*, vol. 10, no. 1, pp. 35–40, 2023.
- Jae Hyun Choi, **Young Ah Shin,** Ik Rae Jeong and Jin Wook Byun, “Analysis of Authentication Protocol Based on Drone Swarm Topology,” *Conference on Information Security and Cryptography-Winter(CISC-W)*, 2022.

– So Yeon Yoon, Hyeon Dong Hwang, Jae Hyun Choi, **Young Ah Shin**, Ik Rae Jeong, and Jin Wook Byun, "Technology Trends and Security Threat Analysis of Drone Docking Stations," *Conference on Information Security and Cryptography-Winter(CISC-W)*, 2021.

– Young Ah Shin, Jae Hyun Choi, Ik Rae Jeong and Jin Wook Byun, "Towards Unmanned Proxy Signature Era : A Design of Unmanned On/Offline Multi-Proxy Signature and Its Implementation," *Conference on Information Security and Cryptography-Winter(CISC-W)*, 2021.

– Jae Hyun Choi, **Young Ah Shin**, Min Seob Lee, and Ik Rae Jeong, "Security Analysis on Authentication/Access Control Protocols for the Internet of Drones," in *Proc. Annual Conference of the Korea Institute of Military Science and Technology(KIMST)*, pp. 1769–1770, Jun. 2021.

– Young Ah Shin, Ik Rae Jeong and Jin Wook Byun, "A Study on Identity-Based Multi-Proxy Signatures and Its Applications to an IoD Authentication Setting," *Conference on Information Security and Cryptography-Summer(CISC-S)*, 2021.

– **Young Ah Shin,** Jae Yeol Jeong and Ik Rae Jeong, "Cancelable Iris Templates Generation Methodology Using Column-Based Slicing," *Conference on Information Security and Cryptography-Summer(CISC-S)*, 2020.

RESEARCH PROJECTS

APAC Cybersecurity Fund Cyber Clinics *Mar. 2026 – Present*

The Asia Foundation, supported by Google.org. Coordinate and manage Korea University Cybersecurity Clinic Center activities, student training, student outreach activities, practical cybersecurity support for MSMEs and community organizations, and participate in regular meetings of the global consortium.

Digital Asset Transaction Tracking to Prevent Malicious Financial Activities *Jul. 2024 – Dec. 2026*

Institute of Information & Communications Technology Planning & Evaluation (IITP), in collaboration with the U.S. Department of Homeland Security. Contribute to research on digital-asset transaction-tracking technologies and security analysis.

University Blockchain Research Initiative (UBRI) *Jun. 2023 – May 2026*

Ripple. Operated a Ripple Validator node, managed the Blockchain Security Research Center at Korea University, and participated in the global XRP Ledger ecosystem, including APEX 2025.

Analysis and Improvement of CBDC Ensuring Both Auditability and User Privacy *Feb. 2025 – Nov. 2025*

Korea Institute of Information Security & Cryptology (KIISC). Investigated privacy-preserving and regulation-compatible tracing mechanisms for CBDC.

Military Cryptography Research Center *Aug. 2021 – May 2024*

Agency for Defense Development (ADD). Designed and evaluated lightweight cryptographic protocols for drone environments and contributed to prototype implementation and performance experiments.

Research on Privacy-Preserving Federated Learning *Jun. 2021 – Feb. 2022*

National Research Foundation of Korea (NRF). Studied privacy-preserving mechanisms for federated learning, including protection of local dataset-size information; research led to an IEEE Access publication.

Security Protocols in Integrated Internet-of-Drones Environments *Mar. 2021 – Feb. 2023*

National Research Foundation of Korea (NRF). Developed identity-based multi-proxy and on/offline cryptographic protocols for resource-constrained drone settings.

ACADEMIC & PROFESSIONAL ACTIVITIES

Peer Review

Reviewer, Proceedings on Privacy Enhancing Technologies (PoPETs) *Jun. 2026 – Present*

Selected as a Tadpole Reviewer (PoPETs'27), an invited role for early-career privacy researchers.

Reviewer, IEEE Access *Jun. 2026*

Organization of Conference, Workshops, or Others

Korea University Cyber Security Clinic Center *Mar. 2026 – Present*

– Develop cybersecurity education and outreach materials, recruit and train 35 undergraduate students from the Division of Smart Security at Korea University.
– Coordinate practical cybersecurity consultations for community organizations (Senior Welfare Center, etc.) and MSMEs, collaborated with Korea Internet & Security Agency (KISA).

Student Representative, Organizing Committee, Korea University International Conference on Cybersecurity and Privacy (KU-ICCSP) 2025 *Oct. 27–28, 2025*

Coordinated logistics and communication with 24 speakers and moderators across eight countries; led an 18-member student operations team and supported diplomatic outreach.

EU Study Visit on Cybersecurity Skills & Higher Education, Korea University *Mar. 12, 2025*

Provided communication support and coordinated campus tours for the international study visit.

SCHOLARSHIPS & AWARDS

Scholarships

University Blockchain Research Initiative (UBRI) *Jun. 2023 – May 2026*

Graduate student research grants of Ripple.

BK21 FOUR (Brain Korea 21 Program for Leading Universities & Students) *Sep. 2020 – Feb. 2026*

Government-funded graduate research scholarship of the Ministry of Education and the National Research Foundation of Korea.

Scholarships from Korea University *2022 – 2025*

2022 General Scholarships, 2022 Research Assistantships.

Scholarships from Sungshin Women's University *2016 – 2019*

2016 Improving the Language Skills, 2016-2019 Sungshin Scholarship, 2017 Sungshin Twenty-First Century Scholarship (full tuition), 2019 Mentoring Scholarship.

Awards

Encouragement Prize & Director's Award, Public Sector Innovation Idea Contest *Dec. 2018*

Ministry of Oceans and Fisheries, Republic of Korea.

COMMUNITY SERVICE

Cybersecurity Education for Older Adults at Naju Senior Welfare Center *Jul. 30, 2026*

Delivered digital-safety training to 70 older adults and staff members, covering online fraud, smishing, and phishing.

PATENTS & SOFTWARE

Registered Patents

- "System and Method for Reputation Management of Payment Channel Hub Based on Blockchain," Registration No. 10-2730269, Nov. 2024.
- "Device and Method for Generating Cancelable Iris Templates Using Column-Based Slicing," Registration No. 10-2664795, May 2024.
- "Unmanned On/Offline Multi-Proxy Signature Method Using Drone," Registration No. 10-2578105, Jul. 2023.

Patent Applications

- "Watchtower-Based Blockchain Payment Channel Hub for System Liveness," Application No. 10-2024-0118130.

– “Bitcoin Transaction Tracking with Known Tapscripts in Bitcoin Taproot Upgrade,” Application No. 10-2024-0117036.
– “Secure Leader Election in Blockchain Multi-Party Payment Channel Hub Based on Group Key Transfer Protocol,” Application No. 10-2023-0116492.
– “Secure Cascade Vertical Federated Learning Against Passive Label Inference Attack,” Application No. 10-2023-0116489.
– “Securing a Local Training Dataset Size in Federated Learning,” Application No. 10-2023-0027255.

Registered Software

– “2024 Drone-User Key Exchange Software Graphical User Interface,” Registration No. C-2024-018129.
– “2024 Drone-User Key Exchange Software Command-Line Interface,” Registration No. C-2024-018128.
– “2023 Authentication and Key Exchange Protocol Implementation Prototype,” Registration No. C-2024-018124.

SKILLS & LANGUAGES

Languages

Korean (Native) | English (Fluent)

Programming

Python (Intermediate), C/C++ (Intermediate), Go (Intermediate), MySQL (Intermediate)